SEPARATEDNESS OF A LIMIT OF SCHEMES WITH SEPARATED TRANSITION MAPS

THE STACKS PROJECT AUTHORS

Editorial note. Source typo kept verbatim: 'O(U $\otimes$ V)' where the argument uses O(U $\cap$ V) (tensor for intersection); recorded, not corrected in the counted span.

Allowed standard prerequisites (editorial).

- Generators of the limit topology. Limits, Section on properties of limits

Source reference. Original reference 086P: sources/stacks-chow/tags.txt, source lines 10399–10399. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01Z6: sources/stacks-chow/tags.txt, source lines 2581–2581. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01KP: sources/stacks-chow/tags.txt, source lines 2058–2058. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01KJ: sources/stacks-chow/tags.txt, source lines 2052–2052. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01T2: sources/stacks-chow/tags.txt, source lines 2359–2359. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01T5: sources/stacks-chow/tags.txt, source lines 2362–2362. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Situation 086P if S is separated, then for some $i_0 \in I$ the schemes S_i for $i \geq i_0$ are separated.*

Proof. Choose a finite affine open covering $S_0 = U_{0,1} \cup \ldots \cup U_{0,m}$. Set $U_{i,j} \subset S_i$ and $U_j \subset S$ equal to the inverse image of $U_{0,j}$. Note that $U_{i,j}$ and U_j are affine. As S is separated the intersections $U_{j_1} \cap U_{j_2}$ are affine. Since $U_{j_1} \cap U_{j_2} = \lim_{i \geq 0} U_{i,j_1} \cap U_{i,j_2}$ we see that $U_{i,j_1} \cap U_{i,j_2}$ is affine for large i by Lemma 01Z6. To show that S_i is separated for large i it now suffices to show that

$$\mathcal{O}_{S_i}(U_{i,j_1}) \otimes_{\mathcal{O}_S(S)} \mathcal{O}_{S_i}(U_{i,j_2}) \longrightarrow \mathcal{O}_{S_i}(U_{i,j_1} \cap U_{i,j_2})$$

is surjective for large i (Schemes, Lemma 01KP).

To get rid of the annoying indices, assume we have affine opens $U, V \subset S_0$ such that $U \cap V$ is affine too. Let $U_i, V_i \subset S_i$, resp. $U, V \subset S$ be the inverse images. We have to show that $\mathcal{O}(U_i) \otimes \mathcal{O}(V_i) \to \mathcal{O}(U_i \cap V_i)$ is surjective for i large enough and we know that $\mathcal{O}(U) \otimes \mathcal{O}(V) \to \mathcal{O}(U \cap V)$ is surjective. Note that $\mathcal{O}(U_0) \otimes \mathcal{O}(V_0) \to \mathcal{O}(U_0 \cap V_0)$

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is of finite type, as the diagonal morphism $S_i \to S_i \times S_i$ is an immersion (Schemes, Lemma 01KJ) hence locally of finite type (Morphisms, Lemmas 01T2 and 01T5). Thus we can choose elements $f_{0,1}, \ldots, f_{0,n} \in \mathcal{O}(U_0 \cap V_0)$ which generate $\mathcal{O}(U_0 \cap V_0)$ over $\mathcal{O}(U_0) \otimes \mathcal{O}(V_0)$. Observe that for $i \geq 0$ the diagram of schemes

$$\begin{array}{ccc} U_i \cap V_i & \longrightarrow & U_i \\ \downarrow & & \downarrow \\ U_0 \cap V_0 & \longrightarrow & U_0 \end{array}$$

is cartesian. Thus we see that the images $f_{i,1}, \ldots, f_{i,n} \in \mathcal{O}(U_i \cap V_i)$ generate $\mathcal{O}(U_i \cap V_i)$ over $\mathcal{O}(U_i) \otimes \mathcal{O}(V_0)$ and a fortiori over $\mathcal{O}(U_i) \otimes \mathcal{O}(V_i)$. By assumption the images $f_1, \ldots, f_n \in \mathcal{O}(U \otimes V)$ are in the image of the map $\mathcal{O}(U) \otimes \mathcal{O}(V) \to \mathcal{O}(U \cap V)$. Since $\mathcal{O}(U) \otimes \mathcal{O}(V) = \operatorname{colim} \mathcal{O}(U_i) \otimes \mathcal{O}(V_i)$ we see that they are in the image of the map at some finite level and the lemma is proved. $\square$